

Z1Power Sugarland BESS

Combined EMS & MBMU Enclosure – Engineering Scope of Work

Prepared for: Electrical Engineer Review & Redesign

1. Background & Project Context

Z1Power previously deployed a CATL EnerOne+ BESS at a Carson, CA site (Site 1). For that project, two separate enclosures were designed and fabricated:

- Indoor EMS Control Enclosure – housing EMS computers, networking, metering, and controls hardware
- Outdoor MBMU/PCS Enclosure – housing MBMU, fire alarm panel, CAN/fire alarm interfaces to CATL battery cabinets, and aux power distribution

The two-enclosure design at Carson was driven by a 400-foot separation between the battery/PCS equipment and the point of interconnection, requiring a separate outdoor panel near the batteries.

The Sugarland, TX site does not have this constraint. All equipment (CATL batteries, Kehua PCS units, metering, and EMS controls) are co-located. The objective for Sugarland is to consolidate both Carson enclosures into a single combined outdoor-rated NEMA enclosure.

2. Sugarland System Overview

Parameter	Detail
Site	Sugarland, TX (ERCOT territory – no SCE requirements)
BESS	375 kW / 1,221 kWh
Battery Units	Three (3) CATL EnerOne+ units (~407 kWh each)
PCS Units	Three (3) Kehua 125 kW grid-forming PCS units
Solar PV	360 kW AC photovoltaic system
Utility Territory	ERCOT (Texas) – no CSIP, no Siemens relay required
Enclosure Type	Single combined outdoor-rated NEMA enclosure (UL508A panel shop)

3. Key Differences from Carson Design

Item	Carson	Sugarland
Number of enclosures	Two (indoor + outdoor)	One (combined outdoor)
Siemens Relay	Required (SCE)	NOT required (ERCOT)
CSIP Gateway	Required (SCE)	NOT required (ERCOT)
CATL Battery Units	Three (3) – Carson was 3-unit config too	Three (3) CATL EnerOne+
PCS Type	Kehua (same)	Kehua 125kW x3 (same)
Equipment separation	~400 ft between battery/PCS and POI	Co-located – no separation
Fiber run	Required between indoor/outdoor panels	Not required (single enclosure)
UL Certification	Required (CA AHJ)	Confirm with TX AHJ

4. Equipment to Include in Sugarland Combined Enclosure

The following equipment from both Carson enclosures shall be consolidated into the single Sugarland enclosure. Items marked EXCLUDE are not required for this site.

4.1 EMS Controls Hardware (from Carson Indoor Enclosure)

Component	Qty	Include?	Notes
Moxa UC-2222A EMS Computer (Site Controller)	2	INCLUDE	Redundant controllers
Moxa EDS-2010-ML Ethernet Switch (LAN)	1	INCLUDE	Internal LAN switch
Moxa EDS-2005-EL Ethernet Switch (WAN)	1	INCLUDE	WAN/internet-facing switch
Mean Well NDR-75-24 Power Supply (24VDC)	2	INCLUDE	Redundant power supplies
Mean Well DRDN20-24 Combiner	1	INCLUDE	Redundant PSU combiner
Phoenix Contact QUINTCAP4KJ DC UPS	1	INCLUDE	EMS hardware backup power
EGauge Revenue-Grade Meter	1	INCLUDE	With CT inputs for PoI and Solar
Potential Transformer	1	INCLUDE	For meter voltage taps
Control Power Transformer	1	INCLUDE	480V to 120/240V aux power
Terminal Blocks (UT6 WH, UT6 PE, UD-80A)	As req.	INCLUDE	Internal wiring terminations
Fuse holders (Altech/Littlefuse KLKR-001)	As req.	INCLUDE	Per fuse schedule
Siemens Relay (grid interconnection relay)	1	EXCLUDE	SCE-specific – NOT required for ERCOT/TX
CSIP Gateway	1	EXCLUDE	SCE-specific – NOT required for ERCOT/TX
Fiber Ethernet Adapter (SFP)	1	EXCLUDE	Only needed for inter-enclosure fiber run – not applicable with single enclosure

4.2 MBMU & Fire Alarm Hardware (from Carson Outdoor/PCS Enclosure)

Component	Qty	Include?	Notes
CATL MBMU (Master Battery Management Unit)	3	INCLUDE	One per CATL EnerOne+ cabinet – 40-pin + 32-pin + 24-pin connectors per Carson wiring
ETH Module (CATL comms interface)	3	INCLUDE	One per CATL cabinet – interfaces with each MBMU
Potter ARC-100 Fire Alarm Control Panel	1	INCLUDE	With CA-6075 Class A Expansion Card
Eaton HT8AEHA	1	INCLUDE	Surface mounted, left side

Strobe/Horn (Abort button)			<i>of panel</i>
Siren	1	INCLUDE	<i>Near top of left side of enclosure</i>
12V Batteries for Fire Alarm panel	2	INCLUDE	<i>Backup power for Potter ARC-100</i>
Moxa IOLogik (DI/DO module)	1	INCLUDE	<i>For alarm/supervisory/trouble contacts</i>
Moxa EDS-2010-ML Ethernet Switch	1	INCLUDE	<i>For MBMU and ETH module connectivity</i>
Mean Well NDR-75-24 PSU (24V for MBMU/IO)	1	INCLUDE	<i>Aux 24V power for MBMU section</i>
480V to 220/380V Aux Power Transformer	1	INCLUDE	<i>Aux power for CATL cabinets (HVAC + BMS)</i>
Distribution breakers (15A, 6A per cabinet)	As req.	INCLUDE	<i>Per Carson BOM – 3 sets for 3 CATL cabinets</i>
Terminal blocks for CAN bus, Fire Alarm SLC	As req.	INCLUDE	<i>USLKG5 series per Carson design</i>
EGauge Meter (outdoor section CT inputs)	1	INCLUDE	<i>BESS CTs – 15 CT inputs per Carson</i>

5. External Field Connections Required

The combined enclosure will interface with the following external equipment via field wiring:

- 3x CATL EnerOne+ battery cabinets – CAN bus, Fire Alarm SLC send/return, 220V HVAC + BMS aux power
- 3x Kehua 125kW PCS units – Modbus/Ethernet comms to EMS
- 480V main bus – voltage taps for meter and aux power transformer
- BESS CTs – wired to EGauge meter CT inputs (up to 15 inputs)
- Solar/PoI CTs – wired to EGauge meter CT inputs
- Internet/WAN connection – from facility network to WAN port of EMS switch
- Protective Earth/Ground bonding

6. Engineering Deliverables Required

The following documents are required from the engineer:

- Updated enclosure wiring schematic (single combined enclosure)
- Component layout drawing showing physical placement within enclosure
- Bill of Materials (BOM) with part numbers and quantities
- Enclosure sizing recommendation (NEMA rating, dimensions)
- Field wiring diagram showing connections to CATL, Kehua PCS, meters, and main bus
- Fabrication-ready package for submission to panel shop for pricing and build

7. Additional Notes for Engineer

- Reference drawings: Z1 Carson Indoor Enclosure (Sheets 1-2) and Z1 Carson PCS Enclosure (Sheets 1-2) are attached as the baseline design
- The EMS software is provided by Battalion Energy – the hardware must be compatible with their Moxa UC-2222A computer and networking architecture
- Enclosure must be outdoor-rated (NEMA 4 minimum) and suitable for Texas climate

- UL508A panel shop fabrication preferred – confirm TX AHJ requirements for UL certification
- The 480V aux power transformer serves the CATL cabinet HVAC and BMS power – three outputs (one per cabinet) at 220/380Y
- Swing panel design from Carson indoor enclosure to be retained if space allows – houses fire alarm panel and MBMU interfaces
- Please provide estimated enclosure dimensions and weight for site planning